

Appendix S4. Additional results

ENA-ABM v0.5 main theoretical experiment

This appendix reports the complete profile-level threshold classification and the prespecified robustness results from the frozen version 0.5 experiment. Appendix S1 describes the model, Appendix S2 documents the preregistered design, and Appendix S3 defines its parameters and configuration. The results below describe the registered theoretical design space; they are not estimates of prevalence or causal effects in real markets.

The accompanying machine-readable tables preserve full numeric precision. Values in the formatted tables are rounded for readability. New adopters exclude seed adopters, and Δ denotes referral minus direct new adoptions under paired simulation.

S4.1 Threshold distribution and bootstrap stability

Of the 96 registered profiles, 91 produced a finite interpretable threshold, four had no competitive referral region within the tested interval, and one remained competitive through the upper boundary $r = 3.00$. No profile exhibited multiple crossings. Among finite thresholds, the median r^* was 1.05, the interquartile range was [1.00, 1.20], and the finite range was [0.85, 2.80]. The median paired-bootstrap interval width was 0.10; 3 profiles (3.1% of all registered profiles) had width greater than 0.50.

Table S4.1. Complete profile-level thresholds, bootstrap intervals, and fixed-cost adoption differences

Profile	Net.	r^*	Class	Bootstrap 95% interval	Δ at 1.00	Δ at 1.20
P001	PA	1.10	finite threshold	[1.05, 1.15]	0.71	-0.44
P002	SW	—	no region	—	-2.02	-4.63
P003	R	0.95	finite threshold	[0.95, 1.00]	-0.14	-2.28
P004	R	1.45	finite threshold	[1.45, 1.50]	2.47	1.17
P005	PA	1.00	finite threshold	[1.00, 1.10]	0.16	-0.24
P006	SW	1.45	finite threshold	[1.35, 1.50]	1.13	0.55
P007	PA	1.00	finite threshold	[0.75, 1.05]	0.21	-0.87
P008	PA	1.05	finite threshold	[1.00, 1.05]	0.27	-0.62
P009	R	1.25	finite threshold	[1.00, 1.35]	0.54	0.14
P010	PA	1.15	finite threshold	[1.10, 1.20]	1.04	-0.20
P011	SW	1.30	finite threshold	[1.00, 1.35]	0.51	0.10
P012	R	1.05	finite threshold	[1.00, 1.15]	0.27	-0.25
P013	SW	1.00	finite threshold	[0.75, 1.00]	0.02	-0.91
P014	PA	1.00	finite threshold	[1.00, 1.05]	0.54	-1.42
P015	R	0.90	finite threshold	[0.90, 0.95]	-0.41	-2.32
P016	R	1.10	finite threshold	[1.05, 1.15]	0.43	-0.17
P017	PA	1.00	finite threshold	[0.75, 1.00]	0.10	-0.86
P018	R	1.20	finite threshold	[1.05, 1.25]	0.38	0.08
P019	PA	0.95	finite threshold	[0.95, 1.00]	-0.14	-1.23
P020	SW	0.90	finite threshold	[0.90, 0.95]	-0.58	-2.79
P021	R	0.95	finite threshold	[0.90, 1.00]	-0.03	-0.65
P022	R	1.40	finite threshold	[1.35, 1.45]	2.52	1.02
P023	PA	1.00	finite threshold	[1.00, 1.05]	0.39	-1.84
P024	PA	1.00	finite threshold	[0.75, 1.00]	0.04	-0.23
P025	PA	1.70	finite threshold	[1.25, 2.00]	0.19	0.12
P026	SW	—	no region	[0.75, 0.75]	-1.41	-3.63
P027	R	1.00	finite threshold	[0.75, 1.00]	0.20	-0.98
P028	R	1.20	finite threshold	[1.10, 1.20]	0.71	0.07
P029	SW	—	no region	—	-1.87	-4.83

Profile	Net.	r^*	Class	Bootstrap 95% interval	Δ at 1.00	Δ at 1.20
P030	PA	1.15	finite threshold	[1.05, 1.25]	0.32	-0.03
P031	PA	1.30	finite threshold	[1.30, 1.35]	3.58	1.09
P032	R	0.95	finite threshold	[0.90, 0.95]	-0.42	-2.62
P033	SW	1.05	finite threshold	[1.00, 1.05]	1.14	-1.56
P034	R	1.05	finite threshold	[1.05, 1.10]	0.70	-0.55
P035	SW	0.95	finite threshold	[0.90, 1.00]	-0.19	-2.35
P036	SW	0.90	finite threshold	[0.90, 1.00]	-0.02	-0.57
P037	R	1.45	finite threshold	[1.40, 1.45]	2.57	1.17
P038	PA	1.00	finite threshold	[0.75, 1.00]	0.09	-0.35
P039	SW	1.50	finite threshold	[1.25, 1.60]	0.98	0.50
P040	SW	1.30	finite threshold	[1.00, 1.30]	0.76	0.10
P041	PA	1.15	finite threshold	[1.10, 1.20]	1.42	-0.10
P042	R	1.00	finite threshold	[1.00, 1.00]	0.28	-0.93
P043	PA	1.00	finite threshold	[0.75, 1.00]	0.13	-1.71
P044	PA	—	no region	—	-2.48	-5.92
P045	SW	1.00	finite threshold	[1.00, 1.05]	0.32	-0.85
P046	SW	1.40	finite threshold	[1.30, 1.50]	0.83	0.39
P047	SW	1.75	finite threshold	[1.50, 2.00]	1.71	1.00
P048	R	1.15	finite threshold	[1.05, 1.20]	0.76	-0.24
P049	R	0.90	finite threshold	[0.90, 0.95]	-0.60	-3.06
P050	R	1.05	finite threshold	[1.05, 1.10]	1.07	-0.98
P051	PA	1.05	finite threshold	[1.00, 1.10]	0.38	-0.43
P052	SW	1.05	finite threshold	[1.00, 1.05]	0.35	-0.60
P053	PA	1.40	finite threshold	[1.00, 1.50]	0.41	0.20
P054	SW	1.35	finite threshold	[1.30, 1.45]	1.11	0.38
P055	SW	0.90	finite threshold	[0.90, 0.90]	-1.28	-5.05
P056	PA	1.00	finite threshold	[0.75, 1.00]	0.06	-0.40
P057	PA	1.05	finite threshold	[1.00, 1.15]	0.24	-0.14
P058	PA	1.05	finite threshold	[1.00, 1.10]	0.71	-0.81
P059	PA	1.40	finite threshold	[1.35, 1.50]	1.70	0.73
P060	R	1.10	finite threshold	[1.10, 1.15]	1.19	-0.59
P061	PA	1.00	finite threshold	[1.00, 1.05]	0.32	-0.68
P062	R	1.05	finite threshold	[1.00, 1.05]	0.35	-0.63
P063	PA	1.05	finite threshold	[1.00, 1.05]	0.73	-1.52
P064	R	0.85	finite threshold	[0.80, 0.90]	-1.20	-3.10
P065	SW	2.50	finite threshold	[2.25, 3.00]	1.85	1.30
P066	R	1.40	finite threshold	[1.30, 1.45]	0.94	0.47
P067	R	1.00	finite threshold	[0.75, 1.00]	0.00	-1.61
P068	SW	0.90	finite threshold	[0.85, 1.00]	-0.12	-0.51
P069	R	1.00	finite threshold	[0.75, 1.00]	0.03	-0.54
P070	R	1.05	finite threshold	[1.00, 1.20]	0.17	-0.08
P071	SW	1.05	finite threshold	[1.00, 1.10]	0.61	-0.99
P072	SW	2.80	finite threshold	[2.00, 3.00]	0.70	0.50
P073	SW	1.05	finite threshold	[1.05, 1.10]	1.24	-1.22
P074	R	0.95	finite threshold	[0.90, 0.95]	-0.47	-2.77
P075	R	1.35	finite threshold	[1.25, 1.35]	0.77	0.24
P076	SW	1.00	finite threshold	[1.00, 1.05]	0.57	-1.32
P077	SW	0.95	finite threshold	[0.90, 0.95]	-0.42	-3.45
P078	R	3.00	right-censored	[2.50, 3.00]	4.41	2.88
P079	SW	0.90	finite threshold	[0.90, 0.95]	-0.86	-3.41
P080	PA	1.00	finite threshold	[0.75, 1.05]	0.00	-0.59
P081	SW	1.00	finite threshold	[1.00, 1.05]	0.33	-0.56
P082	PA	1.00	finite threshold	[1.00, 1.05]	0.34	-1.17

Profile	Net.	r^*	Class	Bootstrap 95% interval	Δ at 1.00	Δ at 1.20
P083	PA	1.05	finite threshold	[1.00, 1.10]	0.27	-0.65
P084	PA	1.15	finite threshold	[1.10, 1.20]	1.03	-0.18
P085	PA	1.00	finite threshold	[0.75, 1.00]	0.03	-1.56
P086	SW	1.00	finite threshold	[1.00, 1.00]	0.36	-1.72
P087	SW	0.95	finite threshold	[0.95, 1.00]	-0.18	-1.00
P088	R	1.15	finite threshold	[1.15, 1.25]	0.86	-0.04
P089	SW	2.55	finite threshold	[2.25, 2.75]	4.54	3.19
P090	R	1.30	finite threshold	[1.00, 1.30]	1.74	0.46
P091	SW	1.15	finite threshold	[1.10, 1.15]	0.79	-0.28
P092	SW	0.90	finite threshold	[0.85, 0.90]	-1.22	-3.39
P093	R	1.00	finite threshold	[0.75, 1.05]	0.10	-1.04
P094	PA	1.05	finite threshold	[1.00, 1.10]	0.54	-0.68
P095	R	1.80	finite threshold	[1.50, 1.90]	3.79	2.32
P096	PA	0.90	finite threshold	[0.90, 0.95]	-0.34	-1.53

Note. PA = preferential attachment; R = ring; SW = small world. The bootstrap interval is the 2.5th–97.5th percentile interval from 500 paired resamples. An absent interval is retained for profiles with no finite bootstrap threshold. The complete machine-readable table also includes alternative threshold definitions, classification flags, bootstrap probabilities, confidence intervals, exposure measures, and conversion measures.

Table S4.2. Threshold distribution by network structure

Network	Profiles	Finite	Median r^*	IQR	Finite range
Preferential Attachment	32	31	1.05	[1.00, 1.12]	[0.90, 1.70]
Ring	32	31	1.05	[1.00, 1.23]	[0.85, 1.80]
Small World	32	29	1.05	[0.95, 1.35]	[0.90, 2.80]

Network medians are similar, but this comparison does not isolate a causal network effect because the profile design varies several factors simultaneously.

S4.2 Fixed-cost comparisons

Table S4.3. Principal outcomes at equal cost and at a 20% referral premium

r	Competitive profiles	Competitive share	Mean Δ	Mean direct adoptions	Mean referral adoptions	Mean $P(Y_R > Y_D)$
1.00	74/96	77.1%	0.49	6.58	7.07	48.0%
1.20	25/96	26.0%	-0.81	6.58	5.77	30.7%

Note. “Competitive” means a nonnegative profile-level mean paired adoption difference. The competitive share describes the 96 deliberately constructed profiles, not the share of firms or markets in a population. Means are averages of the 96 profile-level cell means. The $r=1.20$ values come from the separately documented completion of the preregistered fixed-cost contrast.

At equal attempt cost, referrals were competitive in 74 profiles (77.1%) and produced 0.49 additional adoptions on average. At $r=1.20$, referrals were competitive in 25 profiles (26.0%) and the average difference was -0.81.

S4.3 Exposure-process robustness

Table S4.4. Referral cost thresholds by process regime and network structure

Process regime	Preferential attachment	Ring	Small world
Neutral baseline	1.10	1.25	1.20

Process regime	Preferential attachment	Ring	Small world
Equal high exposure	1.00	1.15	1.10
Referral-exposure advantage	1.60	1.80	1.65
Direct-exposure advantage	No competitive region	0.75	0.65
High acceptance	1.35	1.50	1.45
Low acceptance	No competitive region	No competitive region	No competitive region
Abundant eligible neighbors	1.35	1.60	1.45
No eligible neighbors	No competitive region	No competitive region	No competitive region

Note. Entries are the largest tested cost ratios with a nonnegative mean paired adoption difference. “No competitive region” means that referrals did not reach parity anywhere in the evaluated range. Low acceptance and the absence of eligible neighbors eliminated the competitive region in every network. A referral-exposure advantage increased the finite thresholds, whereas a direct-exposure advantage removed the region in preferential attachment and lowered it in the other networks.

S4.4 Accounting robustness

Table S4.5. Upfront and staged-accounting thresholds

Context	Upfront r^*	Staged r^*	Difference
Baseline	1.15	1.25	0.10
Sociality Low	0.95	1.00	0.05
Sociality High	1.50	1.60	0.10
Budget Scarce	1.10	1.20	0.10
Budget Abundant	1.15	1.40	0.25
Referrers Scarce	1.25	1.30	0.05
Referrers Abundant	1.10	1.15	0.05
Network Ring	1.20	1.25	0.05
Network Preferential Attachment	1.05	1.15	0.10

Note. Staged minus upfront. Staged accounting increased the threshold by a median of 0.10. Eight contexts differed by 0.05 or 0.10; the abundant-budget context was the only one reaching the prespecified material difference of 0.25. The diagnostic stage shares are not empirical cost estimates.

S4.5 Structural robustness

Table S4.6. Thresholds under structural variants

Context	Primary r^*	Conservative r^*	Probability r^*
Baseline	1.20	1.15	1.10
Network Size Small	1.00	0.75	0.75
Network Size Large	1.25	1.00	1.00
Mean Degree Low	1.25	1.00	1.00
Mean Degree High	1.15	1.10	1.10
Rewiring Zero	1.25	1.00	1.00
Rewiring High	1.10	1.05	1.05
Network Ring	1.25	1.00	1.00
Network Preferential Attachment	1.05	1.00	1.00

Note. Structural contexts alter existing model parameters only. Primary thresholds ranged from 1.00 to 1.25. All nine contexts retained a finite, single competitive threshold.

S4.6 Robustness to takeoff and extinction definitions

Table S4.7. Dynamic classifications under alternative definitions

Takeoff share	Quiet ticks	r	Mean Δ	Direct takeoff	Referral takeoff	Direct extinction	Referral extinction
5%	10	1.00	1.51	77.2%	87.8%	22.8%	11.8%
5%	10	1.20	-0.04	77.2%	76.2%	22.8%	23.8%
5%	3	1.00	1.49	74.5%	87.2%	25.5%	12.8%
5%	3	1.20	-0.07	74.5%	72.8%	25.5%	27.3%
5%	5	1.00	1.54	71.8%	86.5%	28.2%	13.5%
5%	5	1.20	0.07	71.8%	75.2%	28.2%	24.8%
10%	10	1.00	1.42	12.5%	28.0%	82.2%	69.0%
10%	10	1.20	-0.01	12.5%	12.2%	82.2%	87.8%
10%	3	1.00	1.43	14.0%	28.2%	86.0%	71.8%
10%	3	1.20	-0.06	14.0%	11.0%	86.0%	89.0%
10%	5	1.00	1.47	12.8%	30.0%	87.2%	70.0%
10%	5	1.20	0.05	12.8%	13.5%	87.2%	86.5%
15%	10	1.00	1.33	0.5%	1.0%	94.2%	91.0%
15%	10	1.20	-0.03	0.5%	0.2%	94.2%	99.8%
15%	3	1.00	1.73	0.2%	1.2%	99.8%	98.8%
15%	3	1.20	0.21	0.2%	0.0%	99.8%	100.0%
15%	5	1.00	1.39	1.0%	0.8%	99.0%	99.2%
15%	5	1.20	-0.01	1.0%	0.0%	99.0%	100.0%
20%	10	1.00	1.51	0.0%	0.0%	92.2%	88.2%
20%	10	1.20	0.03	0.0%	0.0%	92.2%	100.0%
20%	3	1.00	1.14	0.0%	0.0%	100.0%	100.0%
20%	3	1.20	-0.18	0.0%	0.0%	100.0%	100.0%
20%	5	1.00	1.50	0.0%	0.0%	100.0%	100.0%
20%	5	1.20	-0.10	0.0%	0.0%	100.0%	100.0%

Note. Each row contains 400 paired replications of the predefined baseline context. Alternative takeoff shares and quiet periods concern classification of dynamic outcomes rather than a new behavioral mechanism. The mean adoption difference is included as a diagnostic for each separately generated paired cell and should not be interpreted as an effect of changing the takeoff or extinction definition.

S4.7 Machine-readable tables and interpretation boundary

The accompanying data-table package contains the following files:

1. Table_S4_1_Profile_Thresholds_Bootstrap_Fixed_Cost.csv: all 96 profiles, four threshold definitions, classification flags, bootstrap results, and detailed cells at $r = 1.00$ and $r = 1.20$;
2. Table_S4_2_Fixed_Cost_Aggregates.csv: aggregate descriptive comparisons at $r = 1.00$ and $r = 1.20$;
3. Table_S4_3_Process_Robustness.csv: all 24 network-by-process threshold cells;
4. Table_S4_4_Accounting_Robustness.csv: the nine accounting comparisons;
5. Table_S4_5_Structural_Robustness.csv: the nine structural contexts; and
6. Table_S4_6_Dynamic_Definitions.csv: all 24 takeoff/quiet-period classifications.

These tables report outcomes of a reproducible theoretical experiment with diagnostic parameters. They do not establish the empirical frequency of referral advantage, estimate a universal customer-acquisition premium, or justify applying the median threshold to an individual venture without calibration.